Cover Sheet (L4)

Atlantic Technological University

ASSIGNMENT/REPORT COVER SHEET

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I confirm that the work submitted has been produced solely through my own efforts.

Student's signature: **David** Date: **18/10/2026**

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Automated Liquid Cooling and Leak Detection Prototype for Server Racks

by

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L00213709

A report submitted in partial fulfilment of the requirements for a B.Eng (Honours) in Electronic Engineering

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Supervised by: Christopher Crossan/Adam Dunworth

Abstract

Server racks generate a lot of heat in a very small space, and keeping them cool without wasting power is a major challenge in modern data centres. Liquid cooling handles high temperatures much better than air alone, but bringing water near electronics introduces an obvious risk of short circuits if a pipe leaks.

This report covers the design and initial testing of a cooling and monitoring prototype based on the Texas Instruments MSP430G2553 microcontroller. The circuit reads room temperature from a TMP36 sensor and relative humidity from an HIH-5030-001 sensor using the microcontroller's 10-bit ADC. Due to memory space issues, all sensor formulas in the code use integer math instead of floating-point numbers. The main operation consists in the MSP430 generating a 1 kHz PWM signal, fed into a DRI0044 dual motor driver. This setup adjusts the voltage sent to a future 5V water pump across three temperature ranges: off below 22 degrees, half power between 22 and 26 degrees, and full power at 26 degrees or higher. Because standard humidity sensors react too slowly to water spills on the floor. A mechanism was also designed using two exposed wire electrodes and a 10 kOhms pull down resistor. At the moment, the dual-channel ADC readings, UART serial output at 9600 baud, and PWM voltage levels have been tested and confirmed on the oscilloscope. The remaining work for this semester involves adding a LCD screen, an ESP8266/ESP32 WiFi module, and testing the physical water loop.

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Introduction

As servers become more powerful and packed closer together, air conditioning alone is not enough to keep data centres at safe operating temperatures. Many facilities now rely

on liquid cooling loops to pull heat directly away from hardware components. However, it is not optimal for a cooling system to run constantly at 100% power all day. That wastes electricity and wears out the pumps. It needs a control loop that reads the temperature and adjusts the water flow only when necessary. At the same time, the system must watch out for water leaks and condensation. This document presents the first five weeks of work on a data centre cooling and monitoring. Built around the MSP430G2553 LaunchPad, the prototype reads analogue environmental sensors, sends in real time diagnostic data over UART to a PC, and uses Pulse Width Modulation (PWM) to control motor power based on real-time temperature thresholds.

Problem Definition

Running a liquid cooled server rack presents two main problems, energy waste and water damage.

Firstly, a cooling pump that runs at a fixed maximum speed even when the servers are cold, is wasting power and the pump is more likely to fail sooner. Secondly, if a tube breaks or a connector comes loose, water can spill onto live electronics. Relying only on an air humidity sensor like the HIH-5030-001 to catch a leak does not work well in practice. Data centres have strong air conditioning units that dry the air constantly, meaning a puddle can form on the floor long before the room's overall humidity goes up. At the same time, industrial flow meters or commercial leak-sensing cables are very expensive for a small prototype. The goal here is to solve both issues with a low budget using a 16-bit microcontroller. Adjusting pump speed automatically as temperature rises, and catching liquid leaks the exact second water touches the bottom of the tray without running out of microcontroller memory.

Aims and Objectives

Aim

The main goal of this project is to build and test an automated liquid cooling, environmental monitoring, and leak detection prototype for a data centre mock up using the MSP430G2553 microcontroller.

Objective

- **Read Multiple Analog Sensors:** Connect both the TMP36 temperature sensor and the HIH-5030-001 humidity sensor to the MSP430G2553, switching between ADC channels P1.3 and P1.4 in software without signal cross talk.
- **Keep Code Size Small:** Write the sensor conversion math in C using only integer arithmetic, so the compiler does not overflow due to float libraries into the microcontroller's memory.
- **Control Motor Speed with PWM:** Use Timer_A and a DRI0044 motor driver to run a 5V water pump, and an optional fan on the second channel, at 0%, 50%, and 100% duty cycles depending on the measured temperature.
- **Add Direct Water Detection:** Build a simple voltage divider leak probe with a 10 kΩ resistor on pin P1.5 to detect water drops in an isolated tray and cut pump power immediately.
- **Display and Send Data:** Fix UART serial timing at 9600 baud using hardware flags (UCA0TXIFG) for Tera Term logging, later add an LCD screen and Wi-Fi module for cloud monitoring.

Design Brief

The prototype is split into three physical areas for safety: the 3.3V logic side on the LaunchPad, the 5V motor power stage on the DRI0044 driver, and an isolated plastic tray for the possible water tubes leaks. The MSP430G2553 reads room temperature on pin P1.3, air humidity on P1.4, and the leak probe on P1.5. Based on what the sensors read, the C program sets the pump speed. It sends a 1 kHz PWM signal from pin P1.6 and a direction signal from P2.0 into the DRI0044, which then supplies the actual current needed by the motor. While the humidity sensor checks the air for general condensation issues, the two wire probes in the tray act as an emergency switch if water leaks out. Right now, live readings are sent over USB to Tera Term, with an LCD and Wi-Fi connection planned for the second half of the project.

Literature Review

Nowadays server rooms use Computer Room Air Conditioning (CRAC) units to push cold air under raised floors. While air cooling is enough for low power racks, modern high-density servers create local hotspots that air cannot cool down fast enough. Because water carries heat much better than air, direct liquid cooling loops are now widely used in data centres. However, running a cooling pump at 100% speed all day wastes power and wears out the pump. Commercial cooling units solve this by using Pulse Width Modulation (PWM) to slow the pump down when the servers are cold [6].

Putting water pipes near active electronics brings the high risk of water leaks. In commercial data centres, leaks are usually caught using either special sensing cables on the floor, like Raychem TraceTek, or inline flow meters that compare water going in against water coming out. The problem for this small lab prototype is price: commercial sensing cables and their controllers cost hundreds of euros, and cheap Hall effect flow sensors have too much reading error to spot slow drips reliably.

For low-cost microcontroller projects based on the TI MSP-EXP430G2ET LaunchPad [1], checking room conditions with analogue sensors like the Analog Devices TMP36 [2] and the Honeywell HIH-5030-001 [3] is a common approach. As shown in the TMP36 datasheet, this sensor works from a 2.7 V to 5.5 V supply and gives a linear 10 mV/°C output without needing external calibration circuits [2]. Similarly, the HIH-5030-001 runs down to 2.7 V, making both sensors directly compatible with the 3.3 V logic rail of the MSP430G2553 microcontroller [3], [4]. Still, using only an air humidity sensor to catch a water leak on the floor does not work well in a server room, because the room air conditioning dries the air before vapor reaches the sensor. Combining these ambient sensors with a simple two-wire conductive probe, using a 10 kΩ pull-down resistor on a third ADC pin [8], and a motor driver like the DFRobot DRI0044 [5] lets the circuit control pump speed and catch liquid spills immediately at almost zero cost.

Proposed Design and Methodology

System Overview

For safety during lab testing, the circuit is split into three physical parts: the 3.3 V control logic on the LaunchPad [1], the 5 V motor power stage on the driver board, and a separate plastic drip tray that holds the water tubes and leak probe. The general layout is shown in Fig. 1.

The MSP430G2553 microcontroller [4] reads three analogue voltages one after another using its internal 10-bit ADC (ADC10) multiplexer:

- **Pin P1.3 (INCH_3):** Connected to the analogue output of the TMP36 temperature sensor [2].

- **Pin P1.4 (INCH_4):** Connected to the HIH-5030-001 humidity sensor [3].
- **Pin P1.5 (INCH_5):** Connected to the conductive water leak probe. The water leak probe consists in two bare wire ends placed 2 mm apart in the plastic tray with a 10 kΩ pull-down resistor to GND on the breadboard.

Inside the main C loop, the raw ADC numbers are converted into degrees Celsius and relative humidity percentage using integer math. Live values are sent to a PC terminal in Tera Term at 9600 baud over the hardware UART pins (P1.1 and P1.2), using the isolation block jumper connections on the MSP-EXP430G2ET board [1]. For motor control, Timer_A outputs a 1 kHz PWM signal on pin P1.6 while pin P2.0 sets the rotation direction. These two 3.3 V signals feed into PWM1 and DIR1 on the DRI0044 driver [5], which switches the external 5 V supply (VM) to run the water pump on channel 1 (M1+/M1-) and an optional cooling fan on channel 2 (M2+/M2-).

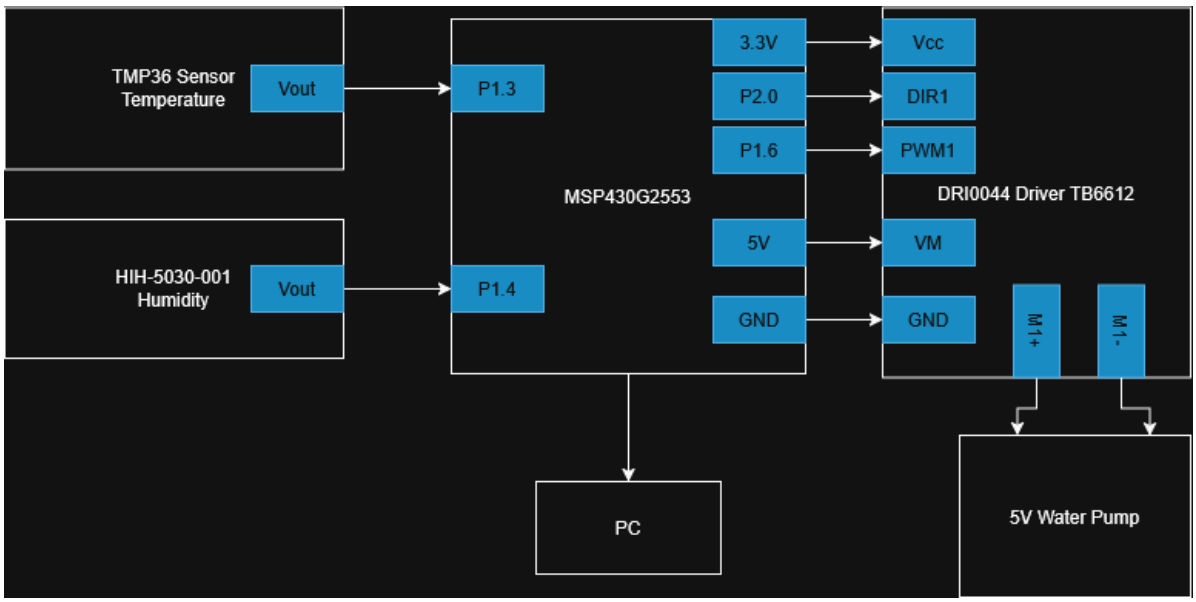


Figure 1: Circuit diagram

Component Selection

Microcontroller MSP430G2553: The main characteristics of this component are its low power consumption, 16-bit RISC architecture, 8-channel 10-bit ADC, hardware UART, and Timer_A PWM outputs [1], [4]. It has enough inputs for the temperature, humidity, and leak sensors. Also generates a 3.3V hardware PWM signal on pin P1.6 without tying up the CPU.

Temperature Sensor (Analogue Devices TMP36): As detailed in its datasheet, the TMP36 operates from a single 2.7 V to 5.5 V supply, requires less than 50 µA of current, and provides a 10 mV/°C scale factor with a 500 mV output at 0°C [2]. This allows direct wiring to the 3.3 V LaunchPad supply and simple conversion in C.

Humidity Sensor (Honeywell HIH-5030-001): This sensor works down to 2.7 V [3], so it shares the same 3.3 V rail as the MSP430G2553 without needing any adjustment.

Motor Driver (DFRobot DRI0044): Its internal MOSFET stage has a much lower voltage drop than older BJT drivers, measuring 4.98 V output from a 5 V supply at 100% duty cycle on the oscilloscope. It also provides a second channel for an auxiliary cooling fan.

Conductive Leak Probe (10 kΩ Pull-Down Resistor and Wire Electrodes): A 10 kΩ resistor holds pin P1.5 at 0 V when the plastic tray is dry. When water touches both wire tips, fluid conductivity completes the circuit to 3.3 V, pulling the ADC voltage up immediately.

A summary of the parts and costs is given in Table 1.

Table 1 – Component List

Item	Quantity	Description	Average Price
MSP-EXP430G2ET	1	TI LaunchPad development board with MSP430G2553 [1], [4]	12.50€
TMP36GT9Z	1	Analog Devices low-voltage temperature sensor (TO-92) [2]	1.80€
HIH-5030-001	1	Honeywell low-voltage relative humidity sensor (SIP-3) [3]	9.50€
DRI0044 Driver	1	DFRobot 2x1.2A DC motor driver (TB6612FNG) [5]	6.90€

5V Water Pump	1	Mini DC 3V–5V submersible cooling pump	3.50€
5V Cooling Fan	1	40x40mm 5V DC brushless axial fan	2.50€
Leak Probe	1	10 kΩ pull-down resistor + 2 exposed wire electrodes	0.20€
16x2 LCD	1	HD44780 character display	4.50€
ESP8266 Module	1	Serial Wi-Fi transceiver board	4.00€

Methodology

The prototype was built and tested in three separate steps so that any wiring mistake on the 5 V motor side would not damage the 3.3 V MSP430G2553 LaunchPad [1].

Step 1: Multi-Channel ADC Reading and Integer Math

The TMP36 temperature sensor was wired to P1.3 (INCH_3) and the HIH-5030-001 humidity sensor to P1.4 (INCH_4). Because the MSP430G2553 uses a single internal sample-and-hold capacitor for its 10-bit ADC, switching channels too quickly caused the humidity pin to copy the temperature voltage when P1.4 was initially left floating. To fix this problem, a change in the Read_ADC function was written. It disables the ADC, selects the new input channel, and waits 16 clock cycles before starting conversion [7]. Instead of using float variables, which quickly overflowed the MSP430G2553 memory, sensor voltages are converted using 32-bit long integer math. For the TMP36 (500 mV offset, 10 mV/°C scale [2]), temperature is calculated from the raw 0–1023 ADC value (adc_t) with a 3.3 V reference:

$$\text{Temperature } (^{\circ}\text{C}) = \frac{(\text{adc_t} \times 330)/1024 - 50}{1}$$

For the HIH-5030-001, applying its 3.3 V transfer function [3] in fixed-point integer math gives:

$$\text{RH } (\%) = \frac{(\text{adc_h} - 155) \times 10}{65}$$

Step 2: Hardware-Polled UART Telemetry

Live values are sent to Tera Term at 9600 baud via P1.1 and P1.2 [1]. At first, using fixed software delays (__delay_cycles(1000)) cut off the second digit of the numbers because 9600 baud take about 1042 µs per character. This was solved by polling the hardware transmit flag before loading each byte into the transmit buffer (UCA0TXBUF).

Step 3: Driver Isolation Check and Oscilloscope PWM Testing

Before wiring the DFRobot DRI0044 driver [5], a cold continuity test was done with a multimeter across VM, VCC, and GND, reading Open Loop (OL) to confirm there were no internal short circuits. Timer_A was set to Up-Mode at 1 MHz SMCLK to output a 1 kHz PWM wave on P1.6 (TA0.1). Heating the TMP36 sensor by hand changed TACCR1 across three levels. The results measured on the oscilloscope across the motor outputs (M1+/M1-) and logic pin P1.6 are shown in Figs. 2, 3, 4 and Table 2.

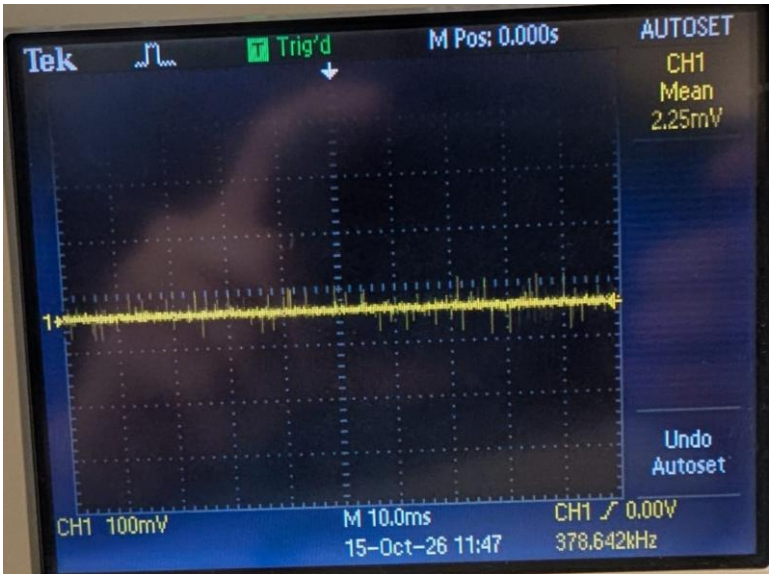


Figure 2: M1+/M1- below 22ºC

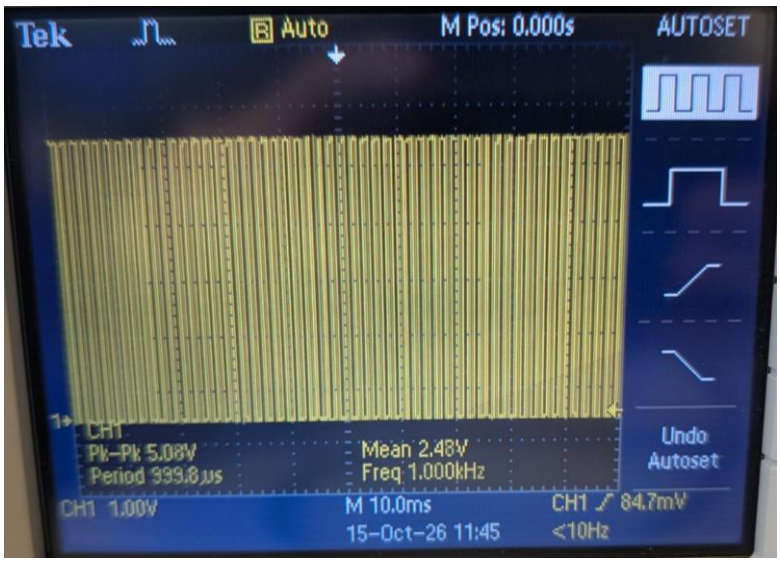


Figure 3: M1+/M1- between 22ºC and 26ºC



Figure 4: M1+/M1- above 26ºC

Table 2 – Oscilloscope PWM Validation Results

Thermal Range	Duty Cycle (TACCR1)	Logic Pin P1.6 (PWM1)	Driver Output (M1+ to M1-)	Observed Waveform
< 22°C	0%	0 V	2.25 mV	Flat DC line
22°C – 25°C	50%	3.3 Vpp (1 kHz)	2.48 V	1 kHz square wave
≥ 26°C	100%	3.3 V DC	4.98 V DC	Flat DC line

Project Plan

The project is divided into a 12-week schedule covering hardware testing, C programming, peripheral expansion, and final enclosure assembly. Tasks from Week 1 to Week 5 are complete, and the remaining weeks focus on the leak detector, LCD display, Wi-Fi link, and closed loop water testing, as shown in Table 3.

Table 3 – Project Timeline

Task	W1-W2	W3	W4	W5	W6	W7-W8	W9-W10	W11-W12	STATUS
Project research									Complete
Component research & pinout study									Complete
TMP36 ADC & UART 9600 baud setup									Complete
Timer_A PWM & driver troubleshooting									Complete
HIH-5030 ADC, UART fix & scope test									Complete
Interim Report submission									Complete
Leak probe & emergency stop									To do
LCD screen wiring & driver code									To do
ESP8266 Wi-Fi cloud telemetry setup									To do
Full water loop, fan & final testing									To do

Risk Assessment and Ethics

Technical and Lab Safety Risks

Working with water next to an active breadboard and a laptop USB port is the biggest safety hazard in this lab setup, since a single drop could easily short-circuit the LaunchPad or the computer. To avoid this, the water pump, the tubing, and the bare wire tips of the leak probe are all kept inside a separate plastic tray away from the electronics, using 20 cm jumper wires to connect back to the dry breadboard. Another hardware issue we already ran into was burning out an earlier motor driver during the first's tests. Because DC motors can draw high startup currents and generate back-EMF voltage spikes when stopping, the replacement DRI0044 board was checked for internal shorts with a multimeter before powering it up, and its 5 V motor supply (VM) is kept separate from the 3.3 V microcontroller rail (VCC). Finally, on the software side, MSP430G2553's memory overflow is a minor risk as the project grows, which was sorted out early on by sticking to integer math so there is enough memory left for the LCD and Wi-Fi code.

Regarding ethics and environmental rules, the circuit only reads room temperature and humidity, meaning no personal data is collected. The main goal of the prototype is to reduce power waste in server racks by using PWM to slow down the cooling pump when the hardware is cold, instead of running motors at full speed all day. All components used in the lab are RoHS compliant, which establish that a product mustn't exceed safe limits for 10 toxic materials.

Progress to Date

- Wired and tested both the TMP36 (P1.3) and HIH-5030-001 (P1.4) sensors on the 3.3 V LaunchPad supply [1], [2], [3].
- Wrote a multi-channel Read_ADC() function in C with a 16-cycle sample-and-hold delay (ADC10SHT_2), fixing the floating-pin crosstalk issue between temperature and humidity readings.
- Replaced fixed UART delays with hardware flag polling (UCA0TXIFG), getting clean serial output on Tera Term at 9600 baud.
- Checked the replacement DRI0044 motor driver [5] with a multimeter and verified the 1 kHz PWM waveforms, P1.6 and M1+/M1-, on the oscilloscope across all three temperature thresholds: <22°C, 22°C–25°C, and ≥26°C.

Next Steps

- **Week 6:** Wire the 10 kΩ pull-down leak detection probe to pin P1.5 and add an if condition in C that forces the pump to shut off and prints a warning when water touches the wire tips.
- **Weeks 7–8:** Connect an LCD in 4-bit mode to display temperature, humidity, and leak status without needing a PC connected.
- **Weeks 9–10:** Interface an ESP8266 Wi-Fi module over serial to send live readings to a web dashboard.
- **Weeks 11–12:** Connect the physical 5 V submersible water pump to channel 1 (M1+/M1-) and the 5 V cooling fan to channel 2 (M2+/M2-) of the DRI0044 for full closed-loop thermal testing.

Bibliography

- [1] Texas Instruments, *MSP-EXP430G2ET LaunchPad Development Kit User's Guide*, SLAU772A, Jul. 2018.
- [2] Analog Devices, *Low Voltage Temperature Sensors TMP35/TMP36/TMP37 Data Sheet*, Rev. H, May 2015.
- [3] Honeywell, *HIH-5030/5031 Series Low Voltage Humidity Sensors Datasheet*, 009042-2-EN, Mar. 2010.
- [4] Texas Instruments, *MSP430G2x53, MSP430G2x13 Mixed Signal Microcontroller Datasheet*, SLAS735J, May 2013.
- [5] DFRobot, *2x1.2A DC Motor Driver (TB6612FNG) SKU: DRI0044 Wiki & Specifications*, 2017.
- *[6] Joyplanes, "How PWM works | Controlling a DC motor with a homemade circuit," YouTube, Feb. 5, 2019. [Online]. Available: http://www.youtube.com/watch?v=5nwNKPs2gco*
- *[7] Jocatronix, "04 Como funciona un ADC de aproximaciones sucesivas," YouTube, Mar. 30, 2023. [Online]. Available: http://www.youtube.com/watch?v=IoQ9Gj3lBvE*
- *[8] Texas Instruments, "Leak Detection, Flow, and Pressure Monitoring in Liquid Cooling Systems," Application Note SNAA422, 2024. [Online]. Available: https://www.ti.com/lit/pdf/snaa422*

Appendices

Datasheets

ANALOG DEVICES

Low Voltage Temperature Sensors

Data Sheet **TMP35/TMP36/TMP37**

FEATURES

Low voltage operation (2.7 V to 5.5 V)
Calibrated directly in °C
10 mV/°C scale factor (20 mV/°C on TMP37)
±2°C accuracy over temperature (typ)
±0.5°C linearity (typ)
Stable with large capacitive loads
Specified −40°C to +125°C, operation to +150°C
Less than 50 µA quiescent current
Shutdown current 0.5 µA max
Low self-heating
Qualified for automotive applications

APPLICATIONS

Environmental control systems
Thermal protection
Industrial process control
Fire alarms
Power system monitors
CPU thermal management

GENERAL DESCRIPTION

The TMP35/TMP36/TMP37 are low voltage, precision centigrade temperature sensors. They provide a voltage output that is linearly proportional to the Celsius (centigrade) temperature. The TMP35/TMP36/TMP37 do not require any external calibration to provide typical accuracies of ±1°C at +25°C and ±2°C over the −40°C to +125°C temperature range.

The low output impedance of the TMP35/TMP36/TMP37 and its linear output and precise calibration simplify interfacing to temperature control circuitry and ADCs. All three devices are intended for single-supply operation from 2.7 V to 5.5 V maximum. The supply current runs well below 50 µA, providing very low self-heating—less than 0.1°C in still air. In addition, a shutdown function is provided to cut the supply current to less than 0.5 µA.

The TMP35 is functionally compatible with the LM35/LM45 and provides a 250 mV output at 25°C. The TMP35 reads temperatures from 10°C to 125°C. The TMP36 is specified from −40°C to +125°C, provides a 750 mV output at 25°C, and operates to 125°C from a single 2.7 V supply. The TMP36 is functionally compatible with the LM50. Both the TMP35 and TMP36 have an output scale factor of 10 mV/°C.

FUNCTIONAL BLOCK DIAGRAM

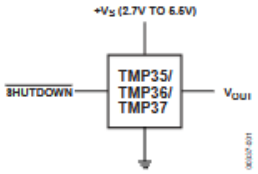


Figure 1.

PIN CONFIGURATIONS

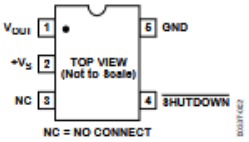


Figure 2. RJ-5 (SOT-23)

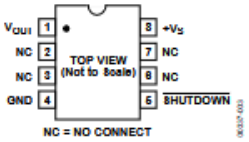


Figure 3. R-8 (SOIC_N)



Figure 4. T-3 (TO-92)

The TMP37 is intended for applications over the range of 5°C to 100°C and provides an output scale factor of 20 mV/°C. The TMP37 provides a 500 mV output at 25°C. Operation extends to 150°C with reduced accuracy for all devices when operating from a 5 V supply.

The TMP35/TMP36/TMP37 are available in low cost 3-lead TO-92, 8-lead SOIC_N, and 5-lead SOT-23 surface-mount packages.

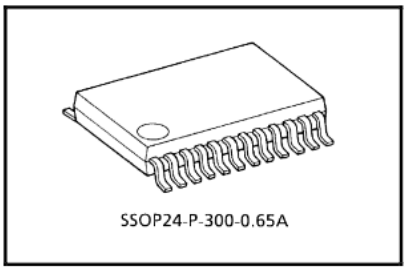
Figure 5: TMP36 Datasheet

Toshiba Bi-CD Integrated Circuit Silicon Monolithic

TB6612FNG

Driver IC for Dual DC motor

TB6612FNG is a driver IC for DC motor with output transistor in LD MOS structure with low ON-resistor. Two input signals, IN1 and IN2, can choose one of four modes such as CW, CCW, short brake, and stop mode.

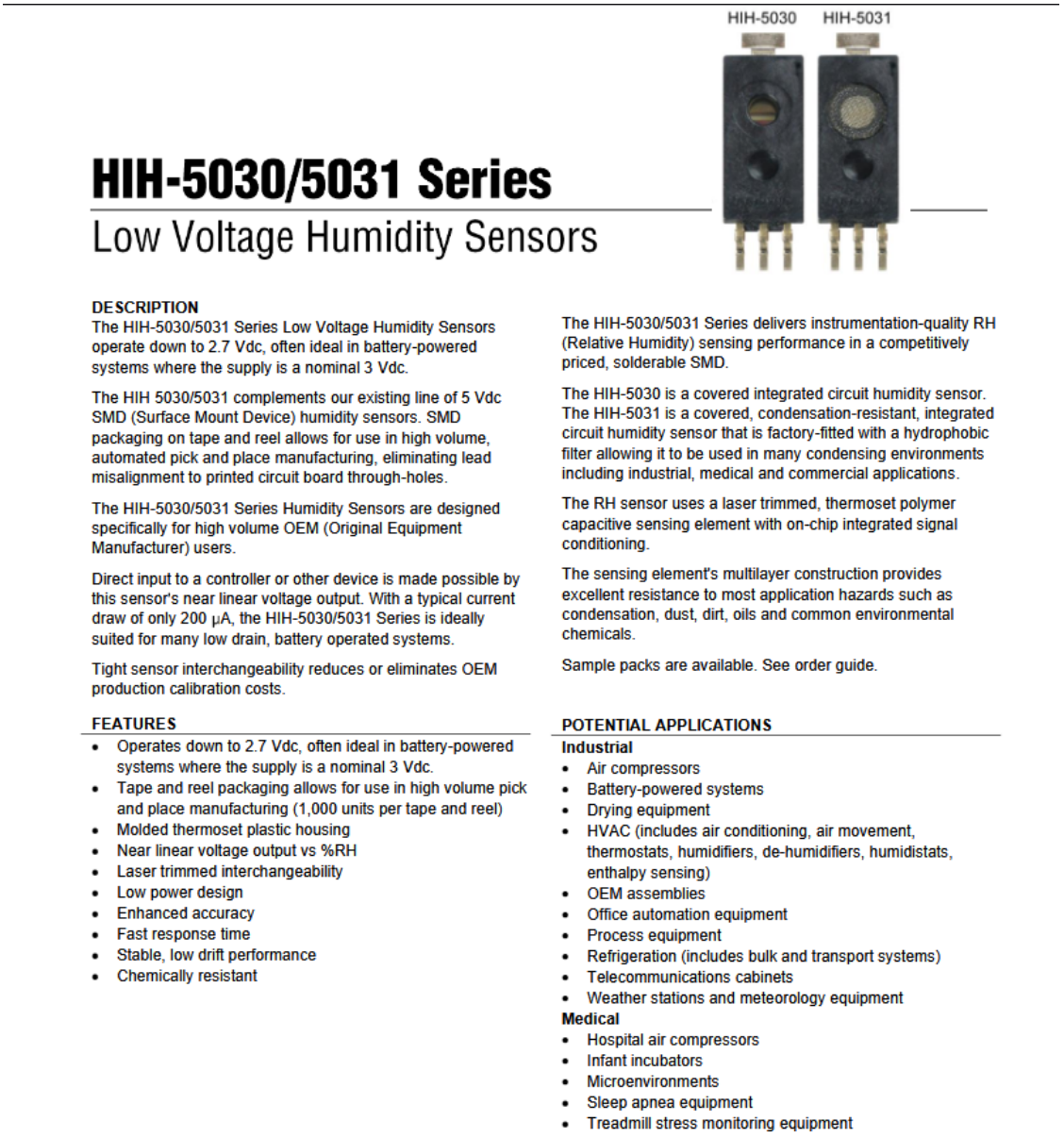


Weight: 0.14 g (typ.)

Features

- Power supply voltage: VM = 15 V(Max)
- Output current: I_{OUT} = 1.2 A(ave)/3.2 A (peak)
- Output low ON resistor: 0.5Ω (upper+lower Typ. @ VM ≥ 5 V)
- Standby (Power save) system
- CW/CCW/short brake/stop function modes
- Built-in thermal shutdown circuit and low voltage detecting circuit
- Small faced package(SSOP24: 0.65 mm Lead pitch)

Figure 6: TB6612FNG datasheet



HIH-5030 HIH-5031

HIH-5030/5031 Series

Low Voltage Humidity Sensors

DESCRIPTION

The HIH-5030/5031 Series Low Voltage Humidity Sensors operate down to 2.7 Vdc, often ideal in battery-powered systems where the supply is a nominal 3 Vdc.

The HIH 5030/5031 complements our existing line of 5 Vdc SMD (Surface Mount Device) humidity sensors. SMD packaging on tape and reel allows for use in high volume, automated pick and place manufacturing, eliminating lead misalignment to printed circuit board through-holes.

The HIH-5030/5031 Series Humidity Sensors are designed specifically for high volume OEM (Original Equipment Manufacturer) users.

Direct input to a controller or other device is made possible by this sensor's near linear voltage output. With a typical current draw of only 200 µA, the HIH-5030/5031 Series is ideally suited for many low drain, battery operated systems.

Tight sensor interchangeability reduces or eliminates OEM production calibration costs.

The HIH-5030/5031 Series delivers instrumentation-quality RH (Relative Humidity) sensing performance in a competitively priced, solderable SMD.

The HIH-5030 is a covered integrated circuit humidity sensor. The HIH-5031 is a covered, condensation-resistant, integrated circuit humidity sensor that is factory-fitted with a hydrophobic filter allowing it to be used in many condensing environments including industrial, medical and commercial applications.

The RH sensor uses a laser trimmed, thermoset polymer capacitive sensing element with on-chip integrated signal conditioning.

The sensing element's multilayer construction provides excellent resistance to most application hazards such as condensation, dust, dirt, oils and common environmental chemicals.

Sample packs are available. See order guide.

FEATURES

- Operates down to 2.7 Vdc, often ideal in battery-powered systems where the supply is a nominal 3 Vdc.
- Tape and reel packaging allows for use in high volume pick and place manufacturing (1,000 units per tape and reel)
- Molded thermoset plastic housing
- Near linear voltage output vs %RH
- Laser trimmed interchangeability
- Low power design
- Enhanced accuracy
- Fast response time
- Stable, low drift performance
- Chemically resistant

POTENTIAL APPLICATIONS

Industrial

- Air compressors
- Battery-powered systems
- Drying equipment
- HVAC (includes air conditioning, air movement, thermostats, humidifiers, de-humidifiers, humidistats, enthalpy sensing)
- OEM assemblies
- Office automation equipment
- Process equipment
- Refrigeration (includes bulk and transport systems)
- Telecommunications cabinets
- Weather stations and meteorology equipment

Medical

- Hospital air compressors
- Infant incubators
- Microenvironments
- Sleep apnea equipment
- Treadmill stress monitoring equipment

Figure 7: HIH-5030/5031 datasheet

Code

```
#include <msp430.h>


#define TXLED BIT0

#define RXLED BIT6

#define TXD BIT2

#define RXD BIT1


void SendData(char *dat);
```

```
unsigned int Read_ADC(unsigned int canal);
void UART_OutUDec(unsigned int n);
int Temperature_value(int ADC_Value);
int Humidity_value(unsigned int ADC_Value);
void Init_Motor_PWM(void);
int temp_C = 0;

char CRLF[] = { 0x0a, 0x0d, 0x00 };
unsigned int ADC_Value;

int main(void)
{
  WDTCTL = WDTPW + WDTHOLD;

  int tmp;

  DCOCTL = 0;
  BCSCTL1 = CALBC1_1MHZ;
  DCOCTL = CALDCO_1MHZ;

  P2DIR |= 0xFF;
  P2OUT &= 0x00;

  P1SEL |= RXD + TXD;
  P1SEL2 |= RXD + TXD;
  P1DIR |= RXLED + TXLED;
  P1OUT &= 0x00;
```

```
UCA0CTL1 |= UCSSEL_2;
UCA0BR0 = 104;
UCA0BR1 = 0x00;
UCA0MCTL = UCBRS2 + UCBRS0;
UCA0CTL1 &= ~UCSWRST;

// Initiate the motor
Init_Motor_PWM();

while (1)
{
  // Read temperature on Pin P1.3
  unsigned int adc_temp = Read_ADC(3);
  tmp = Temperature_value(adc_temp);

  // Read humidity on Pin P1.4
  unsigned int adc_hum = Read_ADC(4);
  int hum = Humidity_value(adc_hum);

  // Send data to Tera Term
  SendData("Temp: ");
  UART_OutUDec(tmp);
  SendData(" C   Humidity: ");
  UART_OutUDec(hum);
  SendData("%");
  SendData(CRLF); // Space jump

  // Motor management , remains the same
```

```
      if (tmp < 22)
      {
         TACCR1 = 0;
      }
      else if (tmp >= 22 && tmp < 26)
      {
         TACCR1 = 500;
      }
      else if (tmp >= 26)
      {
         TACCR1 = 1000;
      }

      __delay_cycles(1000000); // wait 1 sec
   }
}

int Temperature_value(int ADC_Value)
{
   long Vout = (ADC_Value * 3300) / 1023; // Voltage out will be in mV
   int tmp = (Vout - 500) / 10; //Apply offset and divide 10mV/°C
   return tmp;
}

void SendData(char *dat)
{
   unsigned int j = 0;
   while (dat[j] != 0x00)
```

```
  {
    // Polling asking for a free transmission buffer
    while (!(IFG2 & UCA0TXIFG));

    UCA0TXBUF = (dat[j]);
    j++;
  }
}

void UART_OutUDec(unsigned int n)
{
  if (n >= 10)
  {
    UART_OutUDec(n / 10);
    n = n % 10;
  }

  // Polling asking for a free transmission buffer
  while (!(IFG2 & UCA0TXIFG));

  UCA0TXBUF = (n + 0x30);
}

unsigned int Read_ADC(unsigned int canal)
{
  ADC10CTL0 &= ~ENC; // Turn off temporarily the conversion to change the pin

  if (canal == 3)
```

```
  {
    ADC10CTL1 = INCH_3 + ADC10SSEL_3; // Configure channel A3 (Pin 1.3 -
Temperature)
    ADC10AE0 |= BIT3;
  }
  else if (canal == 4)
  {
    ADC10CTL1 = INCH_4 + ADC10SSEL_3; // Configure for channel A4 (Pin 1.4,
humidity)
    ADC10AE0 |= BIT4;
  }

  ADC10CTL0 = ADC10ON + ENC; // Turn on
  ADC10CTL0 |= ADC10SC;     // Start conversion

  while (ADC10CTL1 & ADC10BUSY)
  {
    // wait until its done
  }

  return ADC10MEM; // Return the value measured
}

// Function to read humidity in human values
int Humidity_value(unsigned int ADC_Value)
{
  // Formula in integers based in the datasheet
  int rh = ((ADC_Value - 155) * 10) / 65;
```

```
  // Security limits in case of noise
  if (rh < 0)
    rh = 0;
  if (rh > 100)
    rh = 100;

  return rh;
}

void Init_Motor_PWM(void)
{
  // Configure direction pins (AIN1 y AIN2)
  P2DIR |= BIT0 + BIT2;  // P2.0 y P2.2 as outputs
  P2OUT &= ~(BIT0 + BIT2); // Start with the motor off (LOW)

  // Configure el speed pin (PWMA) in P1.6
  P1DIR |= BIT6;       // P1.6 as output
  P1SEL |= BIT6;       // P1.6 connected to Timer_A

  // Configure Timer_A to generate PWM wave
  TACCR0 = 1000 - 1;// Period of the wave (TACCR0) -> 1000 cycles (1 kHz)

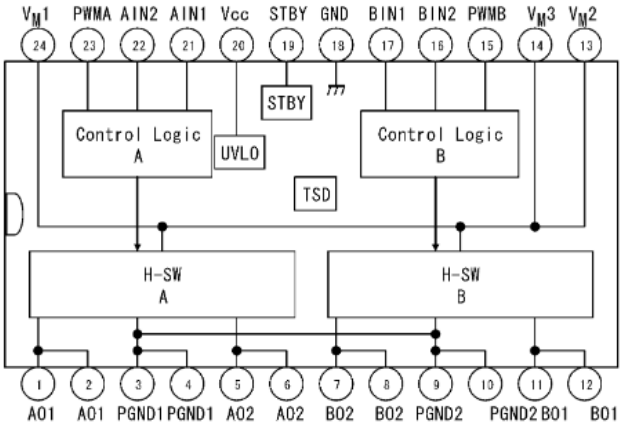
  TACCR1 = 500; // Initial work cycle (TACCR1) -> 500 cycles (50% speed)

  TACCTL1 = OUTMOD_7; // Output mode: Reset/Set, generates the PWM pulse

  TACTL = TASSEL_2 + MC_1; // Turn on the Timer: SMCLK (1MHz), Mode Up (cuenta hasta TACCR0)
```

}

Pinouts



Pin Functions

No.	Pin Name	I/O	Function
1	AO1	O	ch A output 1
2	AO1		
3	PGND1	—	Power GND 1
4	PGND1		
5	AO2	O	ch A output 2
6	AO2		
7	BO2	O	ch B output 2
8	BO2		
9	PGND2	—	Power GND 2
10	PGND2		
11	BO1	O	ch B output 1
12	BO1		
13	VM2	—	Motor supply
14	VM3		
15	PWMB	I	ch B PWM input/200 kΩ pull-down at internal
16	BIN2	I	ch B input 2/200 kΩ pull-down at internal
17	BIN1	I	ch B input 1/200 kΩ pull-down at internal
18	GND	—	Small signal GND
19	STBY	I	"L" = standby/200 kΩ pull-down at internal
20	Vcc	—	Small signal supply
21	AIN1	I	ch A input 1/200 kΩ pull-down at internal
22	AIN2	I	ch A input 2/200 kΩ pull-down at internal
23	PWMA	I	ch A PWM input/200 kΩ pull-down at internal
24	VM1	—	Motor supply

Figure 8: Pinouts of TB6612FNG motor driver

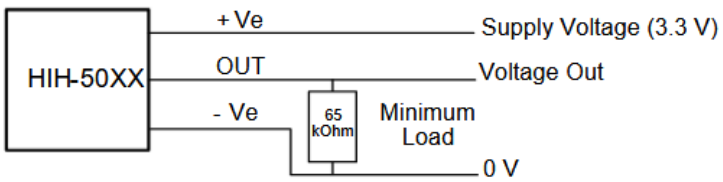


Figure 9: Pinouts of HIH-5030/5031 humidity sensor

2.2.1 MSP430G2553 MCU

The MSP430G2553 is a member of the MSP430 family of ultra-low-power MCUs. MSP430 MCUs features different sets of peripherals targeted for various applications. The MCU architecture, combined with five low-power modes, is optimized to achieve extended battery life in portable measurement applications.

Device features include:

- 1.8-V to 3.6-V operation
- 16-bit RISC architecture up to 16-MHz system clock
- 16KB of flash memory and 512 bytes of SRAM
- 8-channel 10-bit ADC
- 8-channel comparator
- Two 16-bit timers with three capture/compare registers (Timer_A)
- 24 GPIOs
- One universal serial communication interface (USCI_A) supports UART, IrDA, and SPI
- One USCI (USCI_B) supports SPI and I^2C

Figure 4 shows the pinout of the MSP430G2553 20-pin N (PDIP) package.

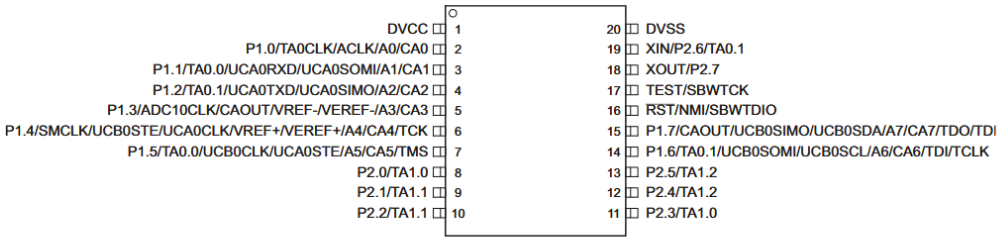


Figure 4. MSP430G2553 20-Pin N Package (Top View)

Figure 10: Pinouts of MSP430G2553

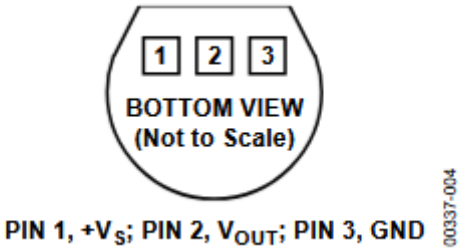


Figure 11: Pinouts of TMP36

Diagrams

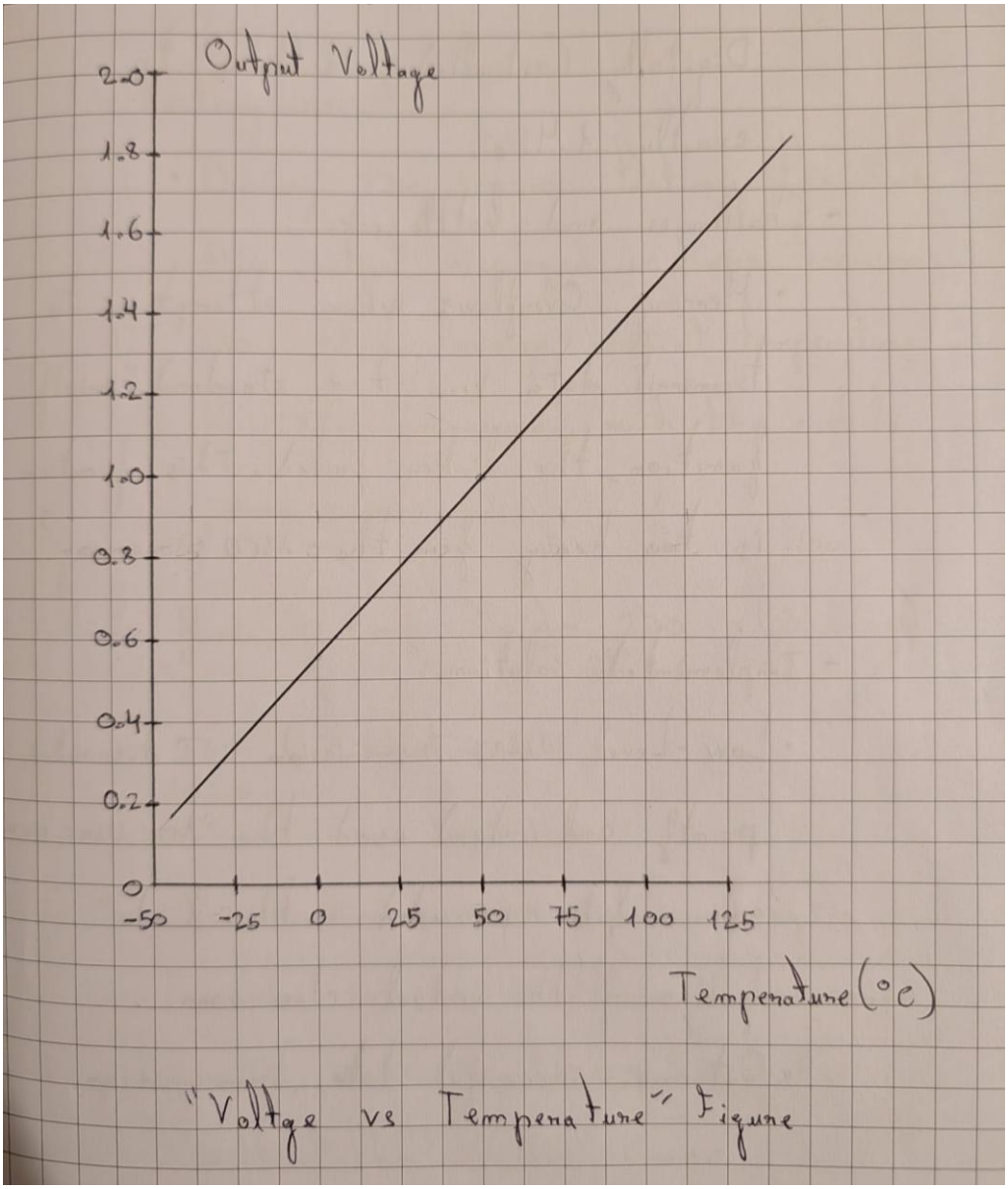


Figure 12: Voltage vs Temperature Figure from TMP36

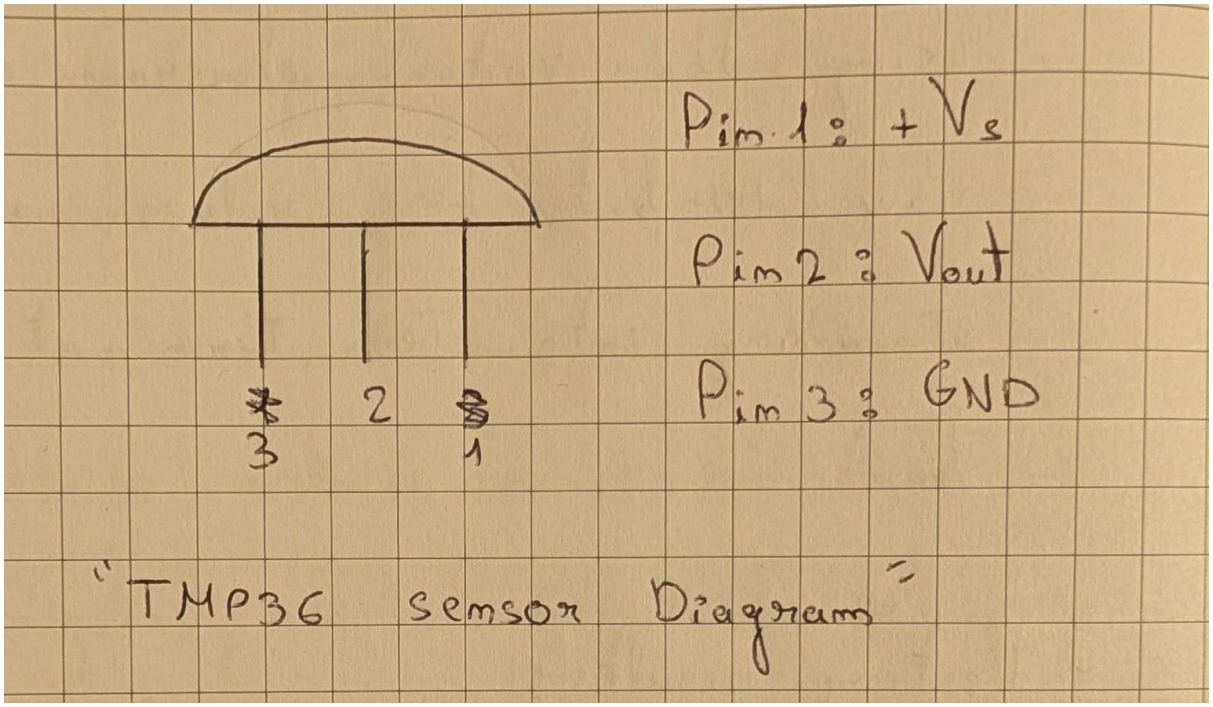


Figure 13: TMP36 Pinouts diagram

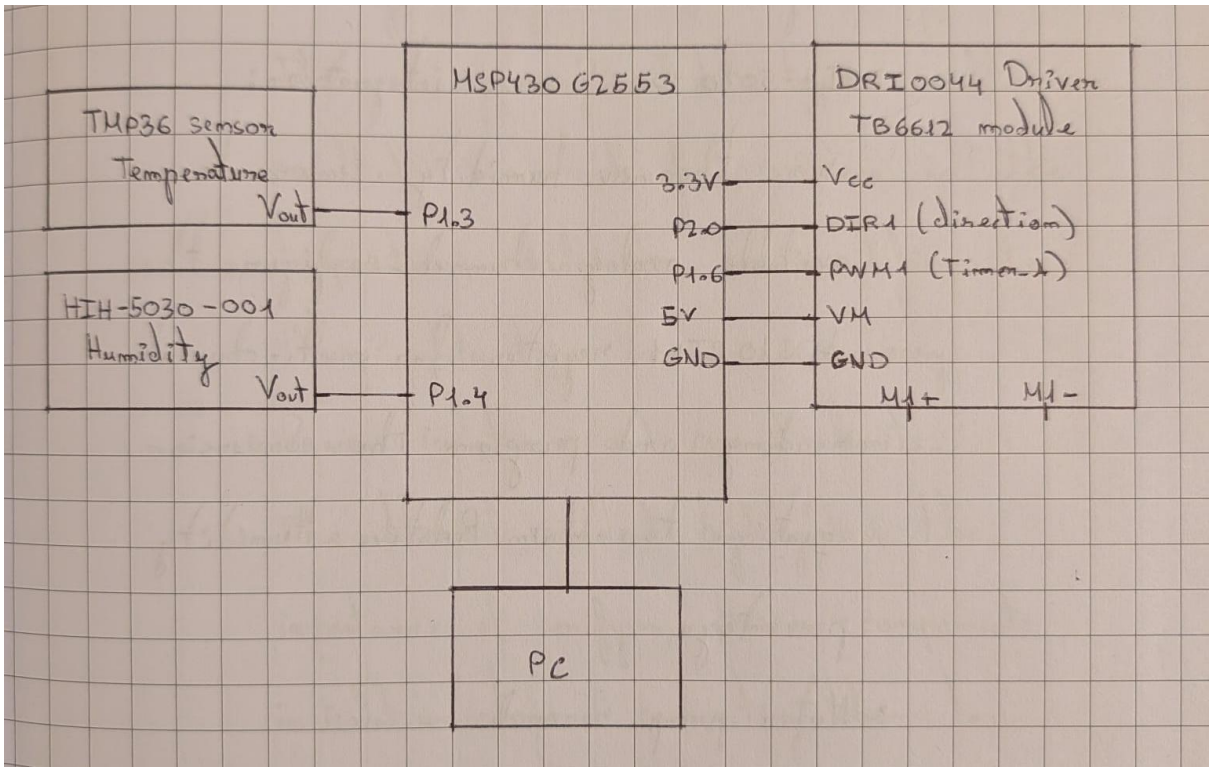


Figure 14: Diagram of the current components

Schematics

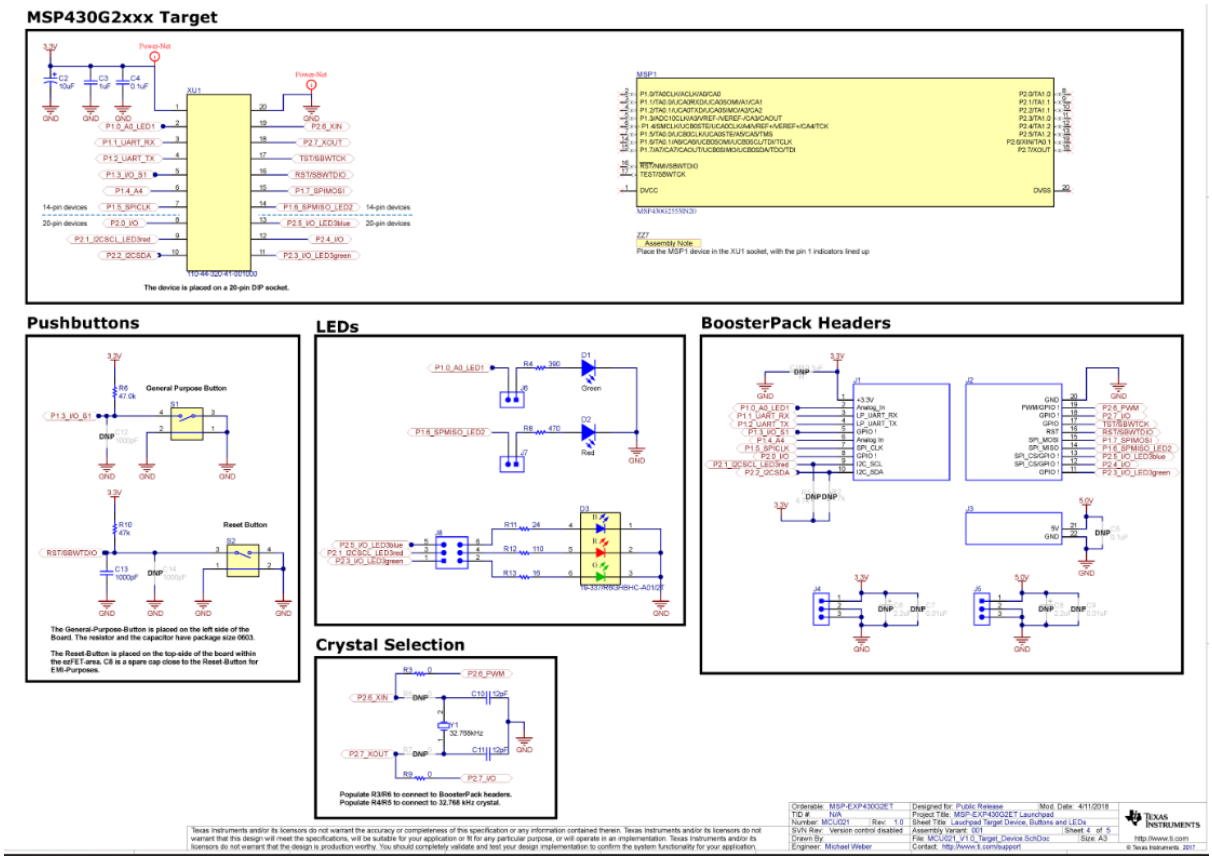
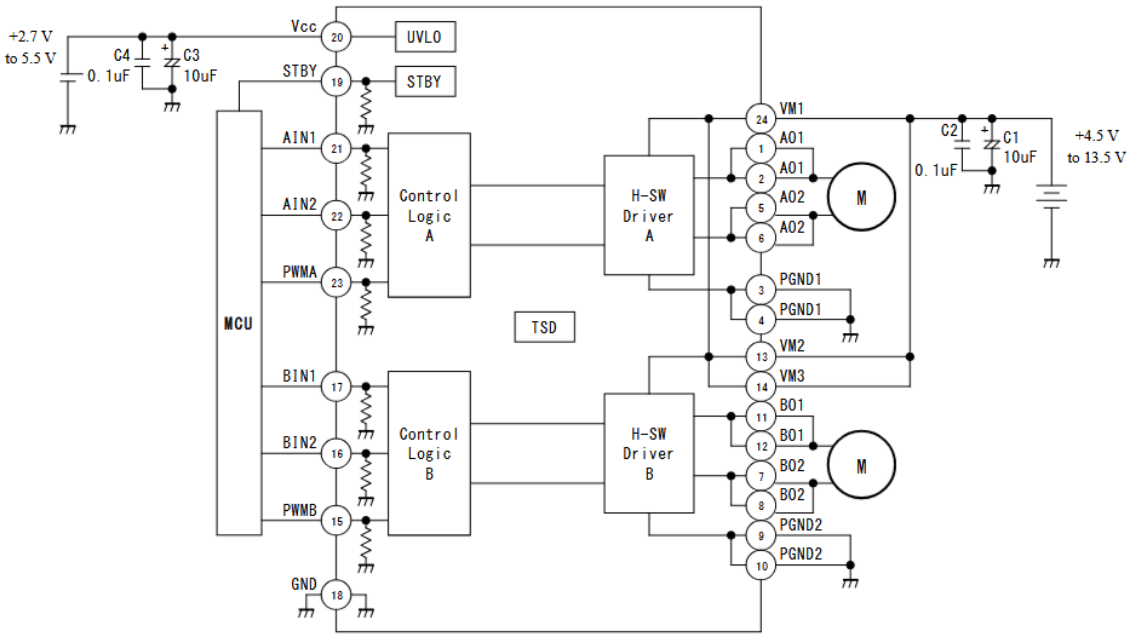


Figure 15: MSP-EXP430G2ET Schematic

Typical Application Diagram



Note: Condensers for noise absorption (C1, C2, C3, and C4) should be connected as close as possible to the IC.

Figure 16: TB6612FNG Schematic

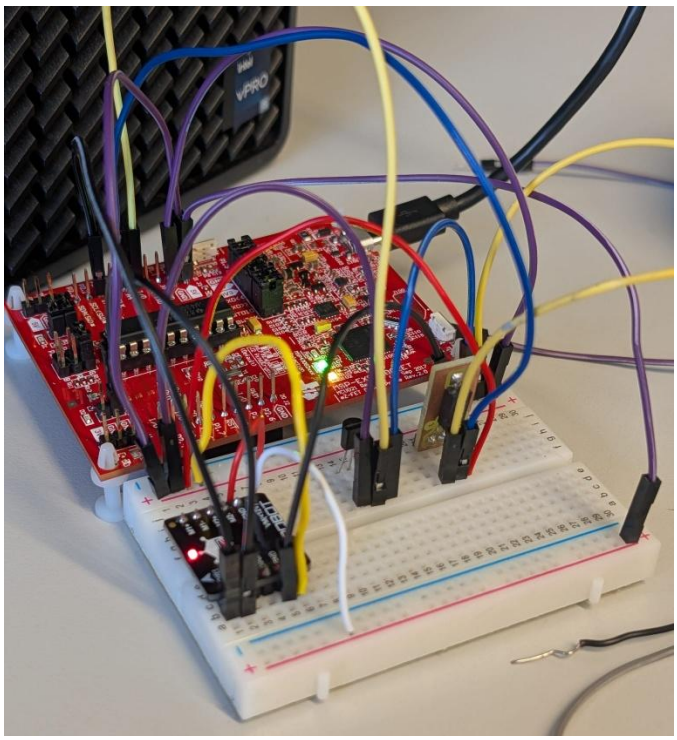


Figure 17: Circuit mounted